

Geometry-based Methods in Vision

CMU 16-822, Fall 2024

Instructor: Shubham Tulsiani

Administrivia

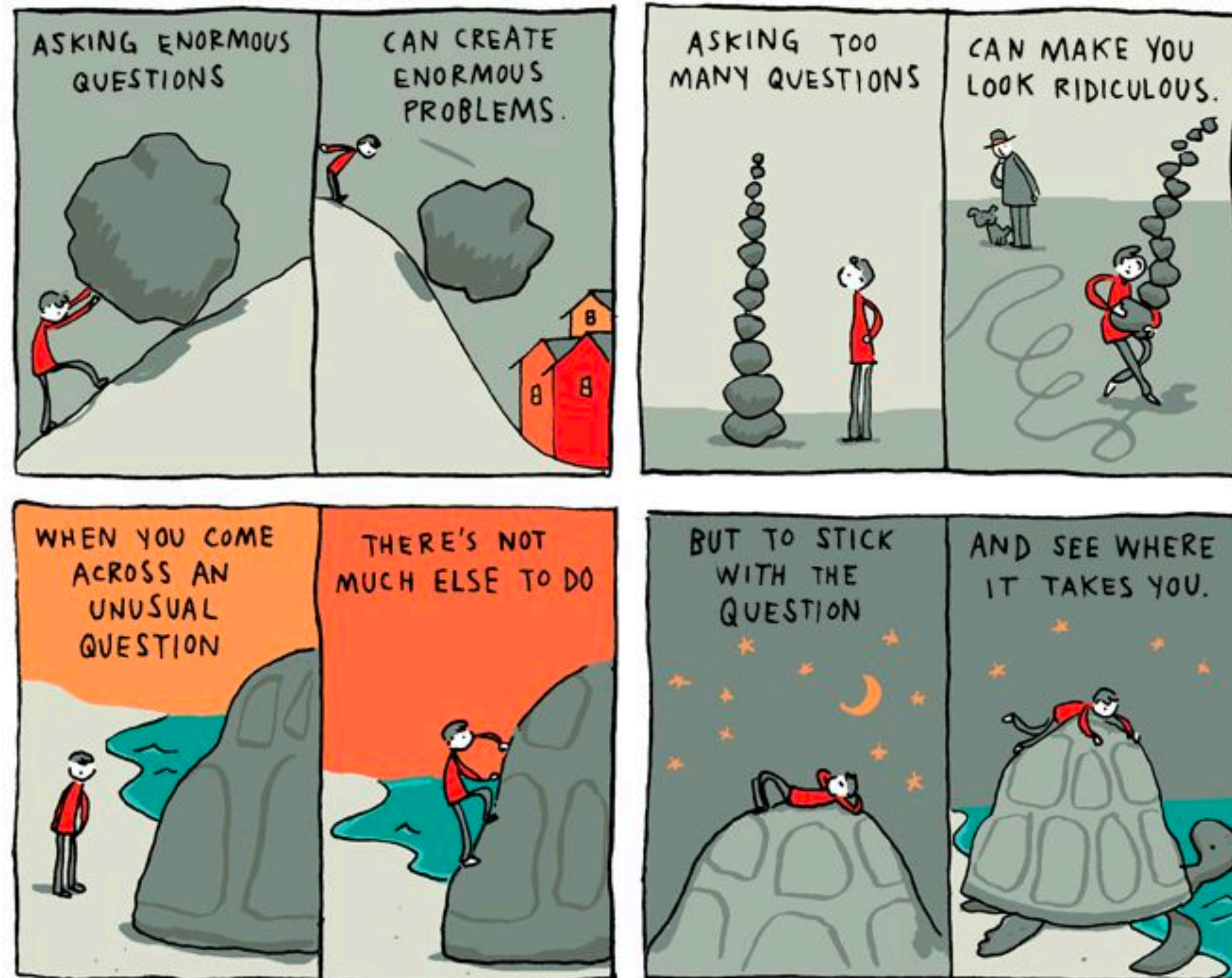
- A0 and PS0 out
 - *Due on Thu*
 - *Please let us know if you have issues creating webpages*
 - *Please self-report attendance when submitting PS0*

Lecture 3: 2D Projective Geometry (Transforms & Rectification)



Figure courtesy: Hartley & Zisserman

A reminder to ask questions!



GRANT SNIDER

2D Projective Geometry: Lines and Points

$\mathbb{P}^2$ Projective 2D space represented by (x, y, z) s.t. $(x, y, z) \equiv (\lambda x, \lambda y, \lambda z)$

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} u \\ v \\ 0 \end{bmatrix} \equiv \begin{bmatrix} \frac{1}{\lambda}x + u \\ \frac{1}{\lambda}y + v \\ \frac{1}{\lambda} \end{bmatrix} \rightarrow \begin{bmatrix} u \\ v \\ 0 \end{bmatrix}$$

Line Equation: $\mathbf{l}^T \mathbf{x} = 0$

Line $\mathbf{l}$ intersecting points $\mathbf{x}_1$ and $\mathbf{x}_2$

$$\mathbf{l} = \mathbf{x}_1 \times \mathbf{x}_2$$

All points at infinity lie on a line

$$\mathbf{l}_\infty = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Intersection point $\mathbf{x}$ of lines $\mathbf{l}_1$ and $\mathbf{l}_2$

$$\mathbf{x} = \mathbf{l}_1 \times \mathbf{l}_2$$

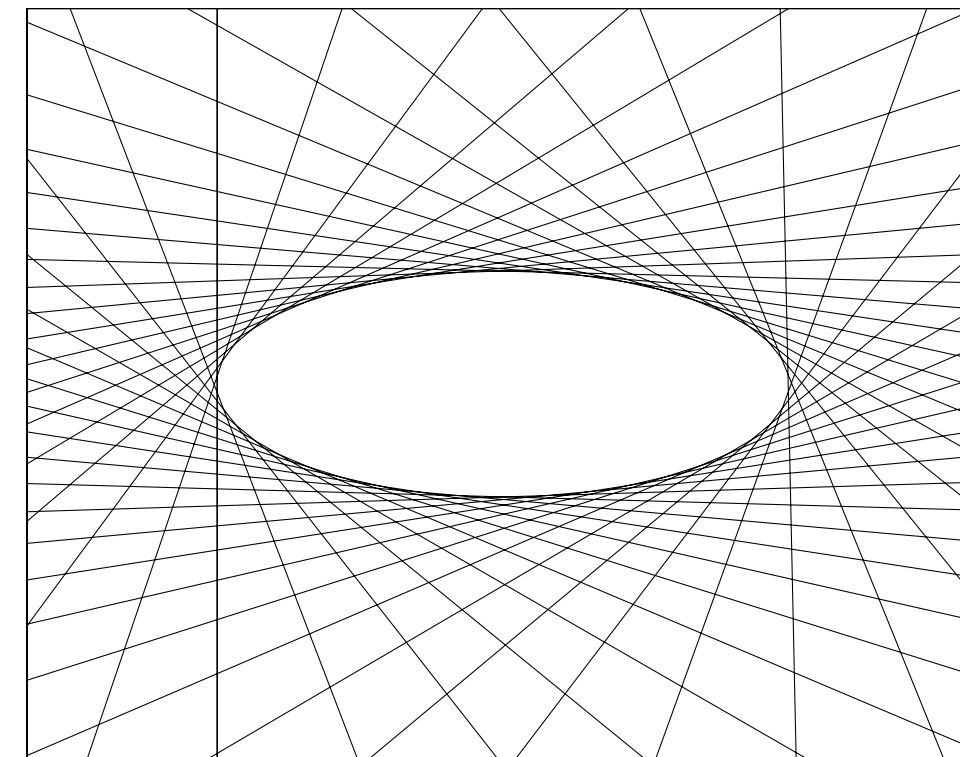
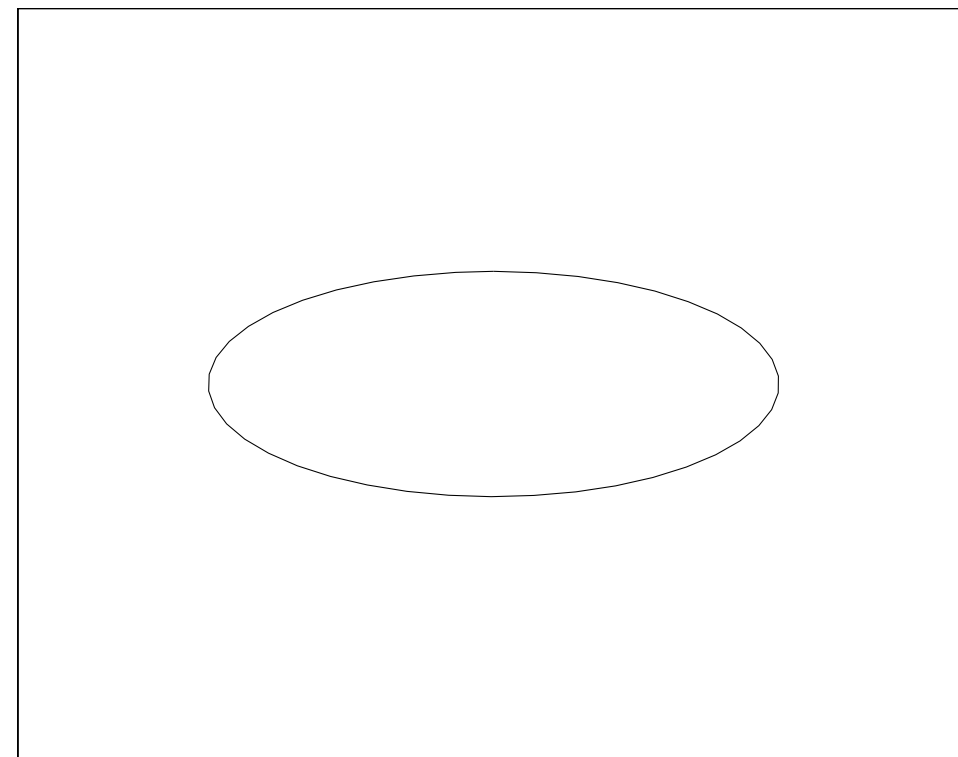
2D Projective Geometry: Conics

$$\mathbf{x}^T C \mathbf{x} = 0$$

point conic

$$\mathbf{l}^T C^* \mathbf{l} = 0$$

line conic



$$C^* = C^{-1} \text{ (up to scale)}$$

(adjoint of C if $\det(C) = 0$)

Transformations in Projective Space

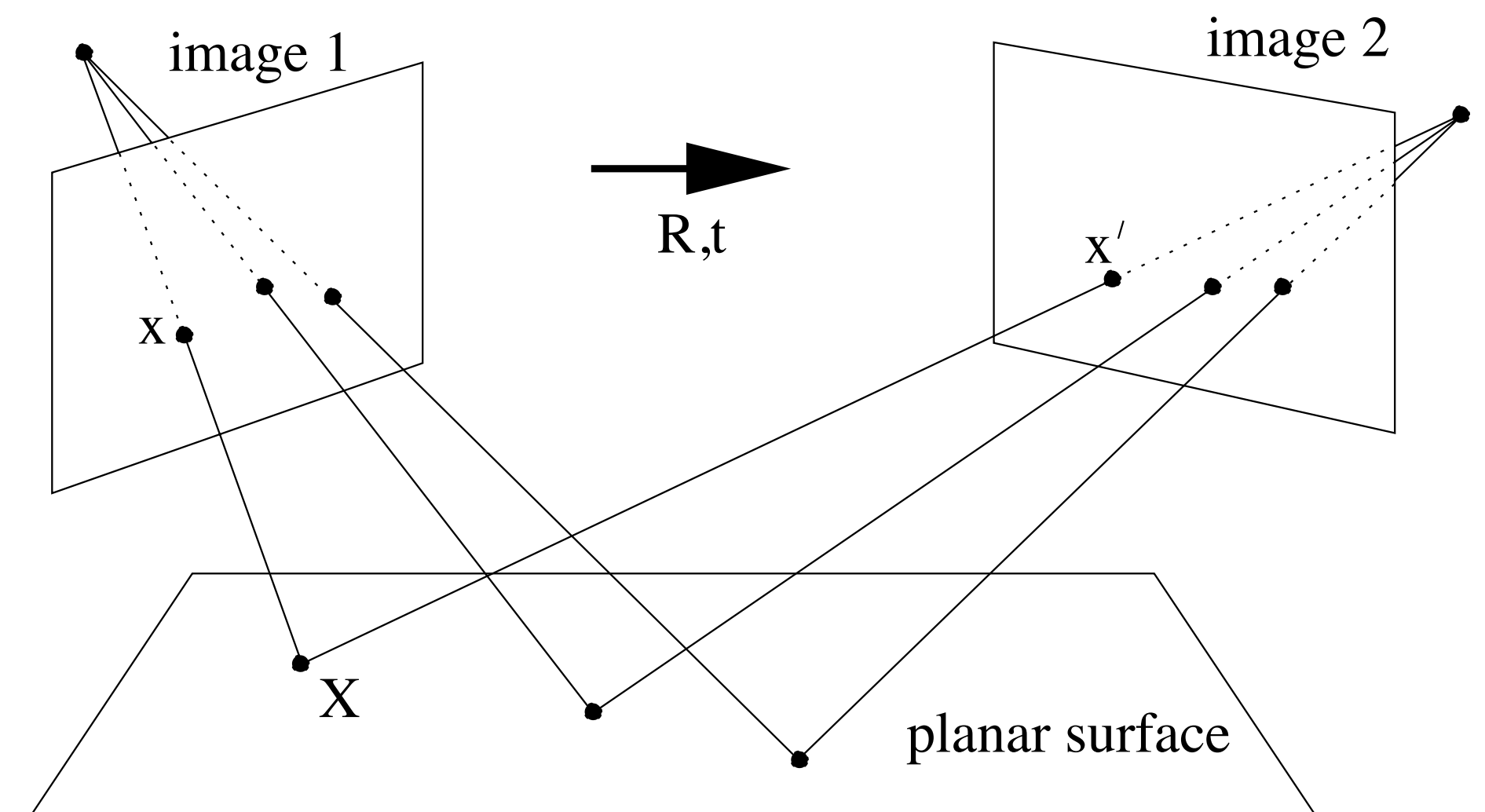
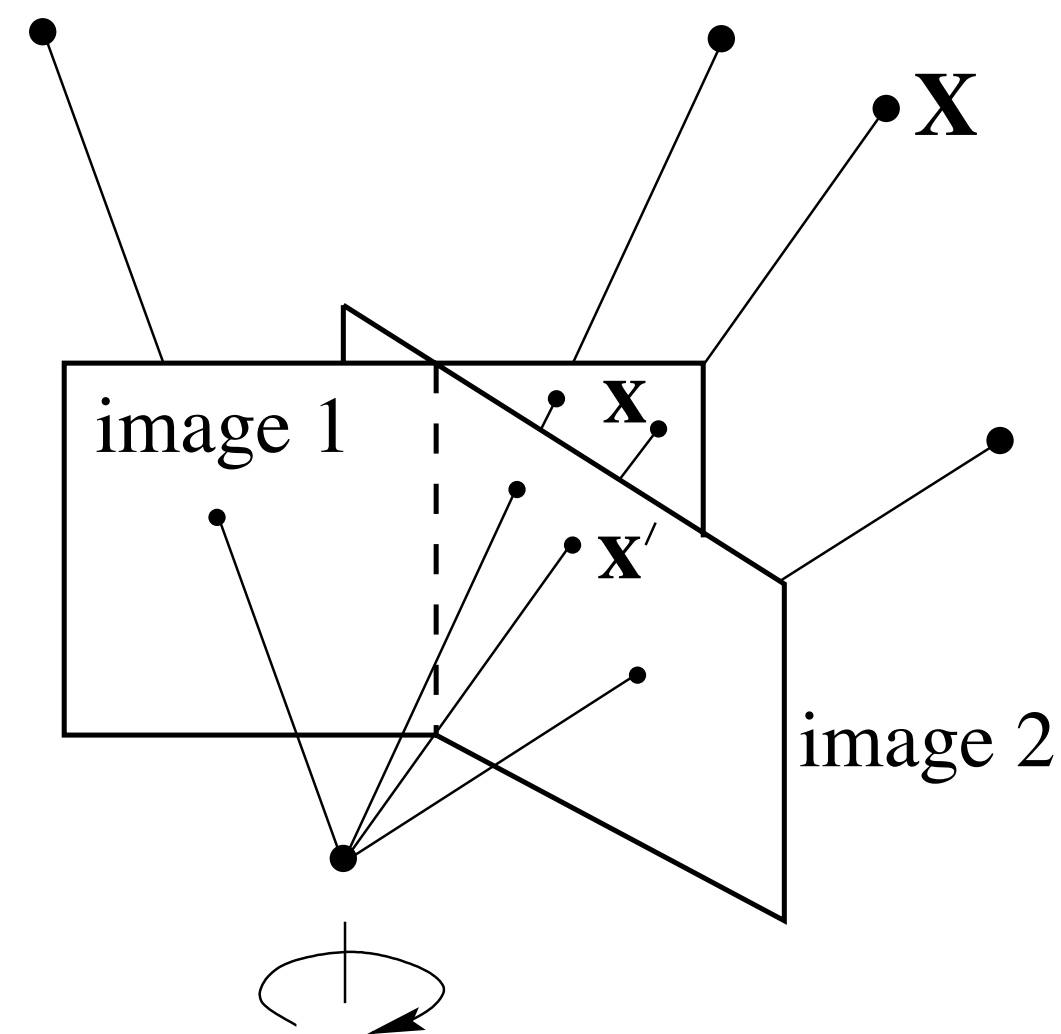
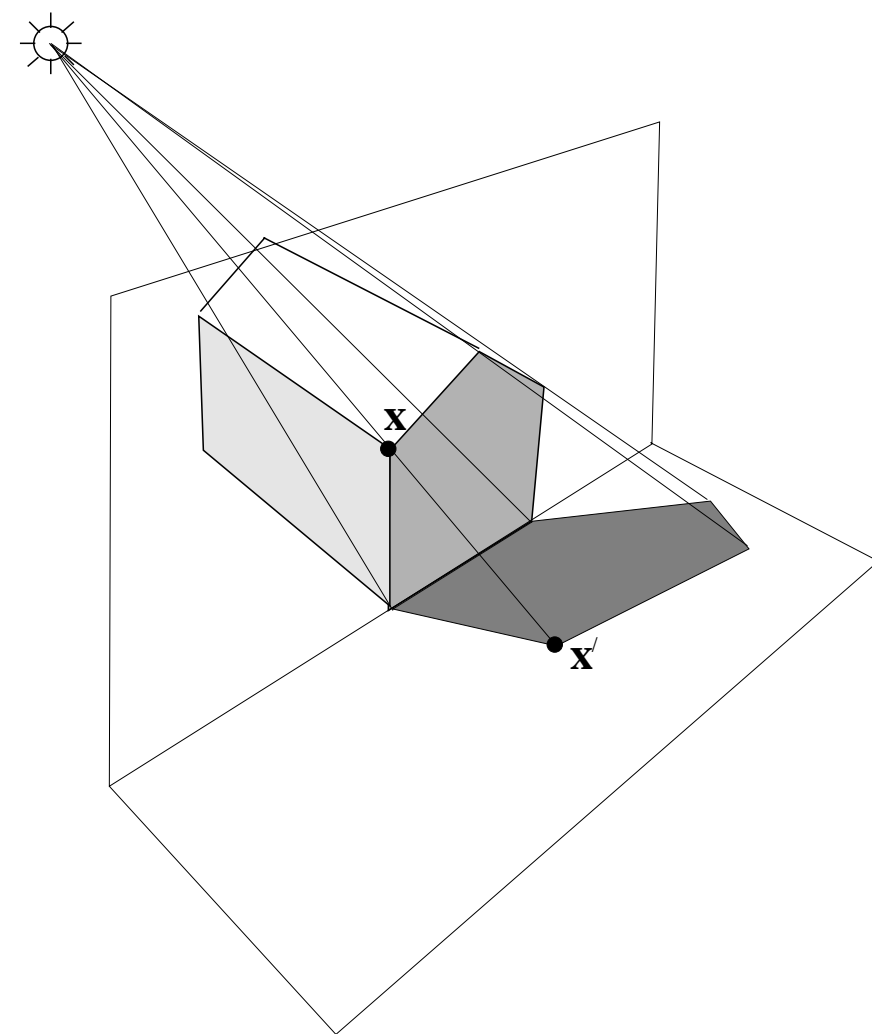
$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$



A *projectivity* is an invertible mapping h from $\mathbb{P}^2$ to itself such that three points $\mathbf{x}_1, \mathbf{x}_2$ and $\mathbf{x}_3$ lie on the same line if and only if $h(\mathbf{x}_1), h(\mathbf{x}_2)$ and $h(\mathbf{x}_3)$ do.

Transformations in Projective Space

$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$



All these 2D projective transforms have an underlying 3D structure that gives rise to them!

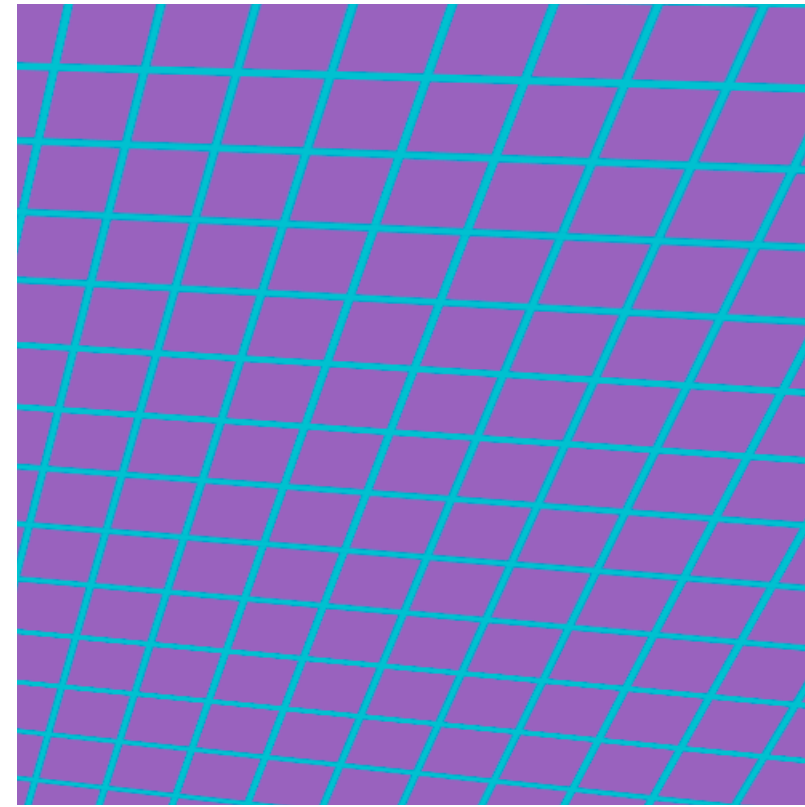
Transformations in Projective Space

$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$

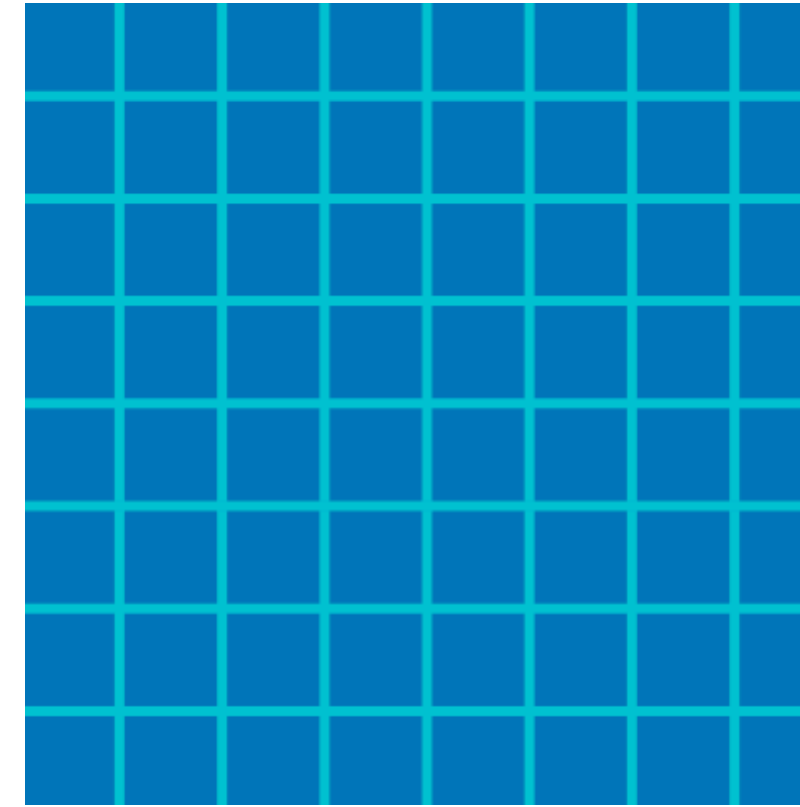
Projectivity/Homography: **H** is an invertible matrix

Q: How many degrees of freedom for **H** ?

Transformations in Projective Space

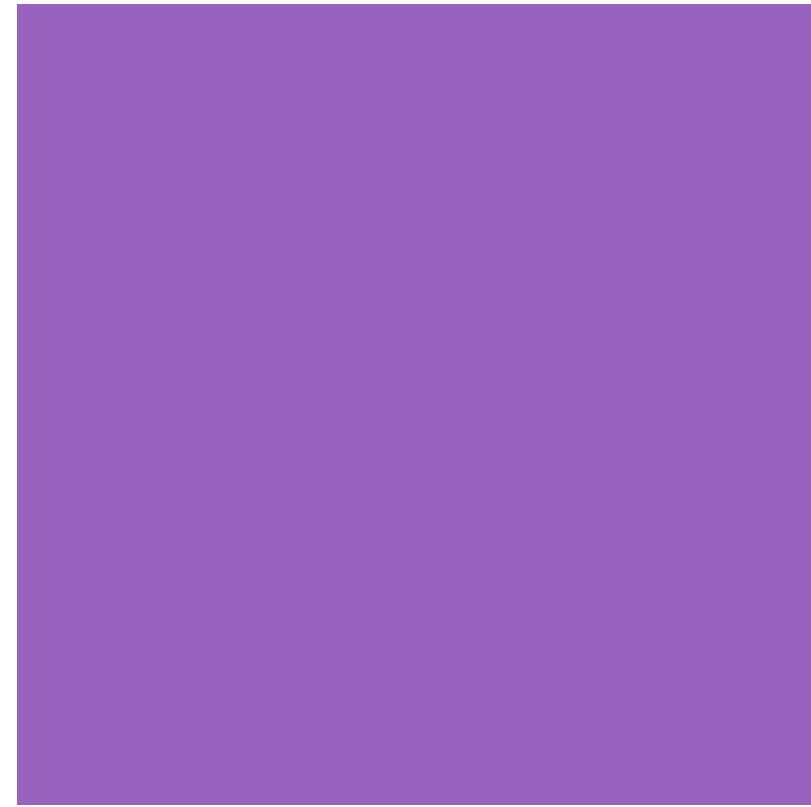


$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$



Note: the images visualize associations between $(x, y, 1)$ and $(x'/w', y'/w', 1)$

Transformations in Projective Space



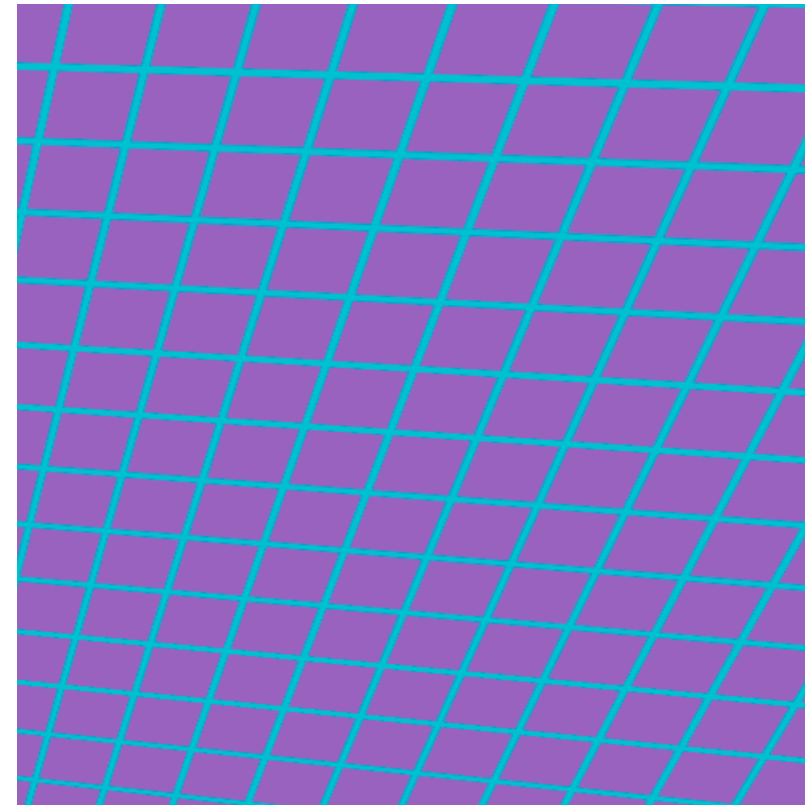
$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$



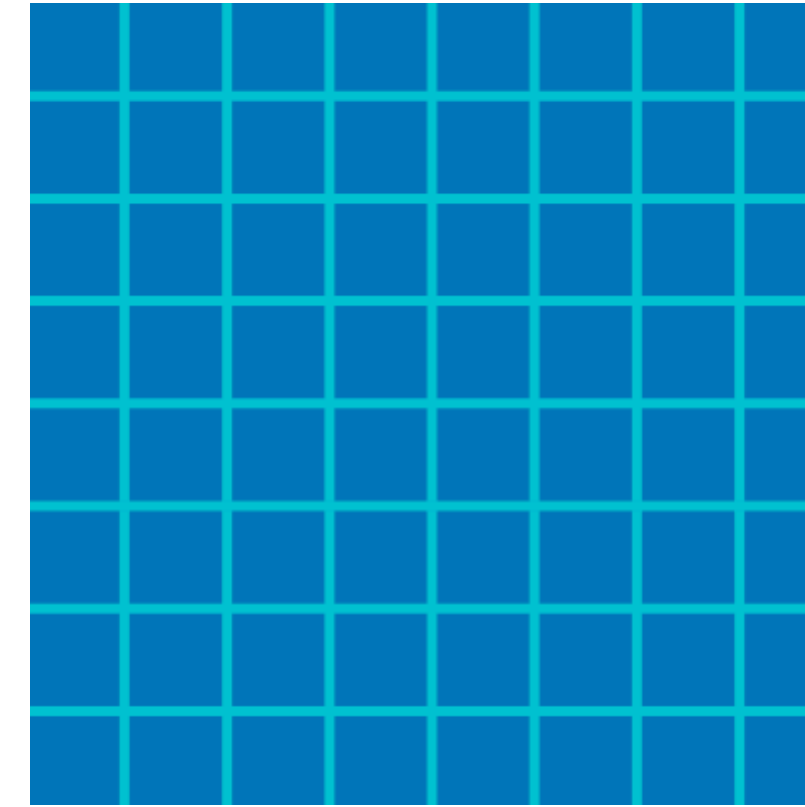
Both, the source and transformed domains are the same i.e. $\mathbb{P}^2$

There is nothing ‘special’ about the points/lines in the source (blue) domain

Transformations in Projective Space



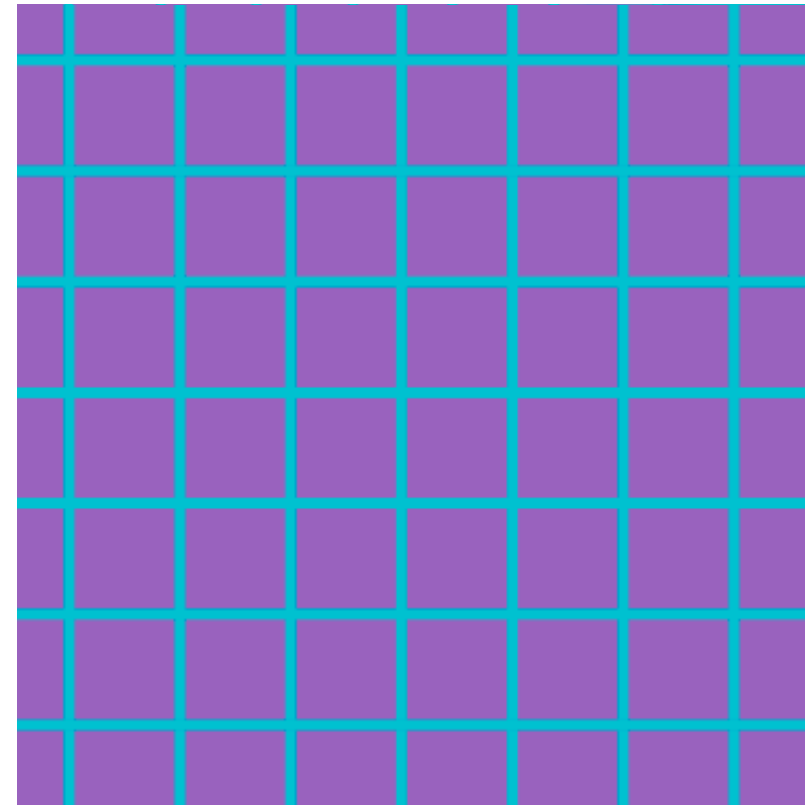
$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$



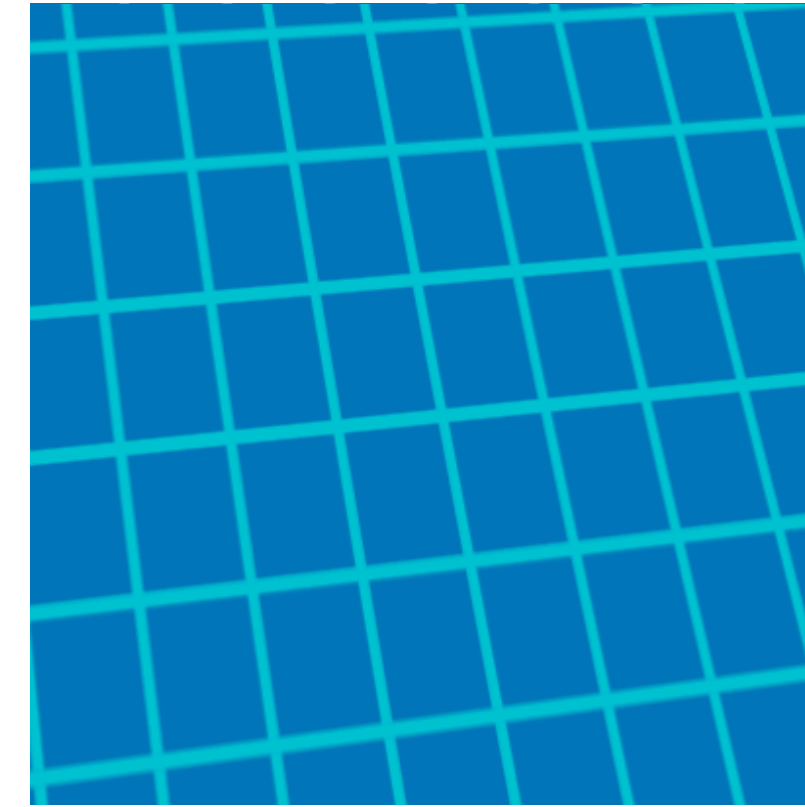
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Transformations in Projective Space



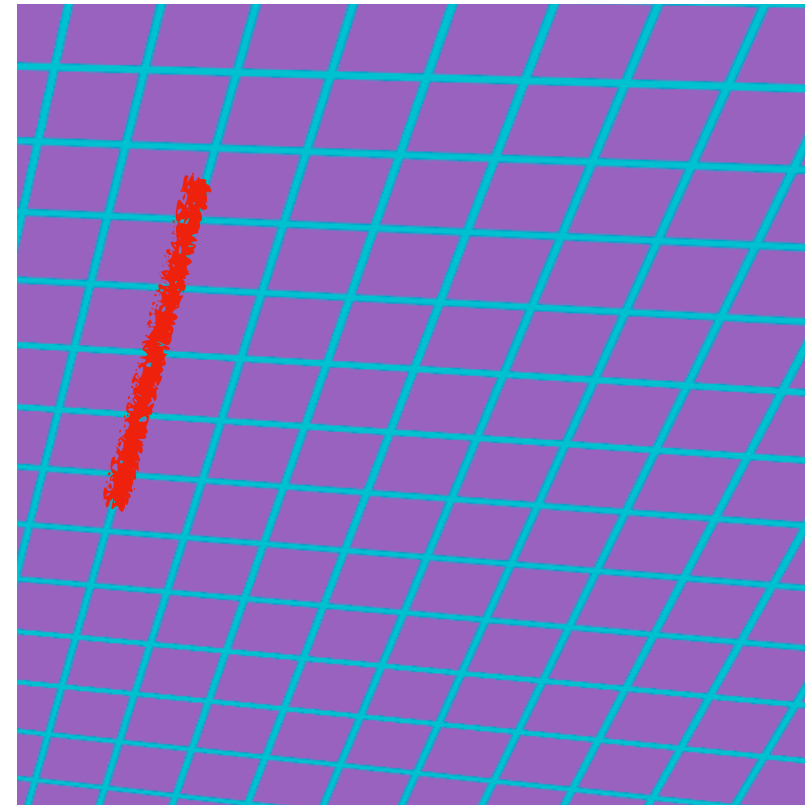
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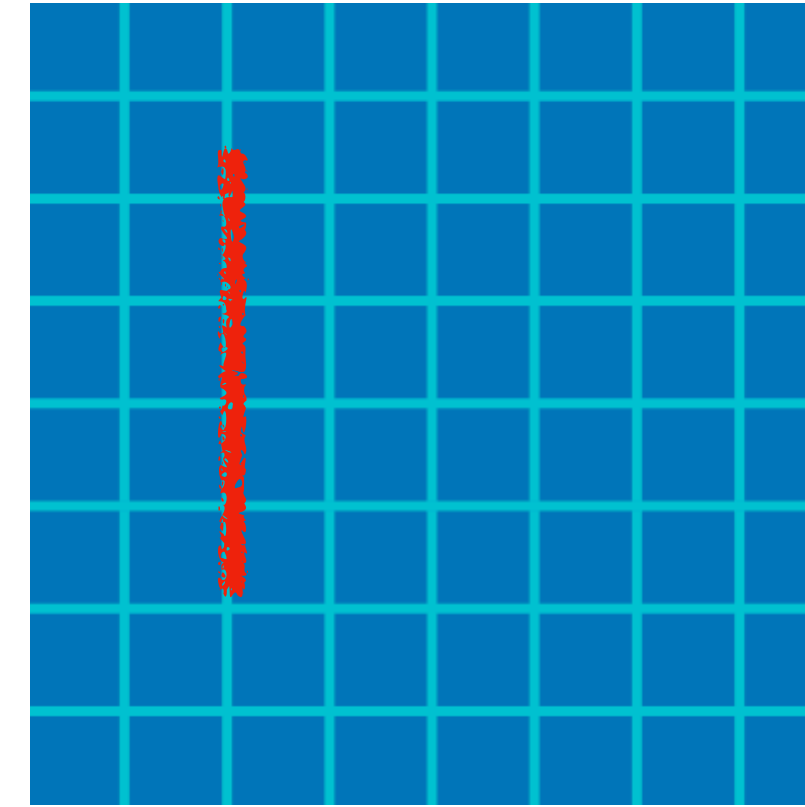
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Transformations in Projective Space



$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$



Q: How does a line $\mathbf{l}$ transform under a homography?

Proof:

$$\mathbf{l}' = \mathbf{H}^{-\top} \mathbf{l}$$

$$\mathbf{l}^\top \mathbf{x} = 0$$

$$\mathbf{l}^\top \mathbf{H}^{-1} \mathbf{H} \mathbf{x} = 0$$

$$(\mathbf{H}^{-\top} \mathbf{l})^\top \mathbf{H} \mathbf{x} = 0$$

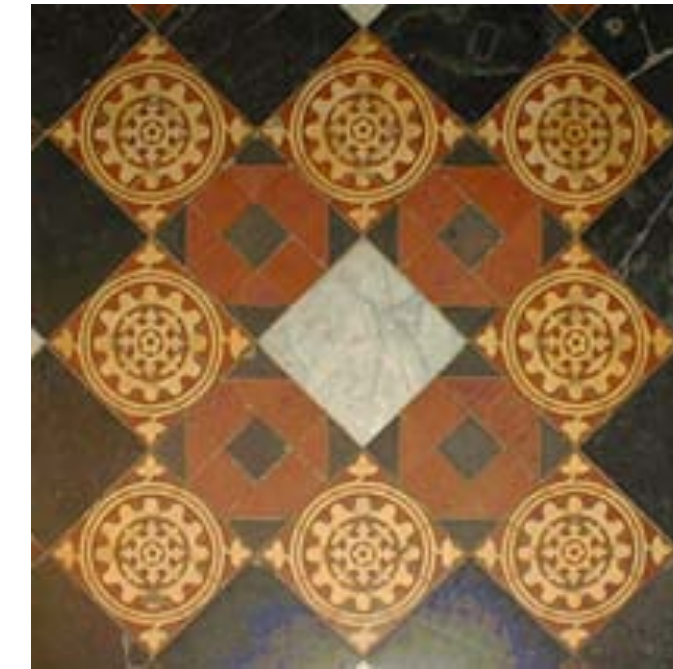
Can similarly show:

$$\mathbf{c}' = \mathbf{H}^{-\top} \mathbf{c} \mathbf{H}^{-1}$$

A hierarchy of transformations: Similarity



$$\mathbf{x}' = \mathbf{H}_s \mathbf{x} = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \mathbf{x}$$



Q: How many degrees of freedom?

~~3 DoFs — 1 for s , 1 for R , 2 for $\mathbf{t}$. And -1 for scale ambiguity.~~

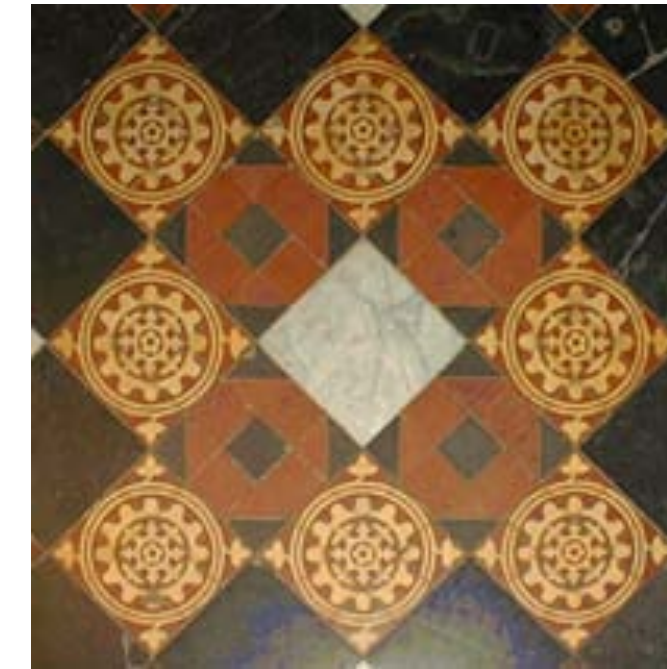
4 DoFs — 1 for s , 1 for R , 2 for $\mathbf{t}$

Preserves angles between lines

A hierarchy of transformations: Affine



$$\mathbf{x}' = \mathbf{H}_A \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \mathbf{x}$$



Q: How many degrees of freedom?

$$4 + 2$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

where $\mathbf{K}$ is upper-triangular, with $\det(\mathbf{K})=1$

Preserves ‘parallelism’

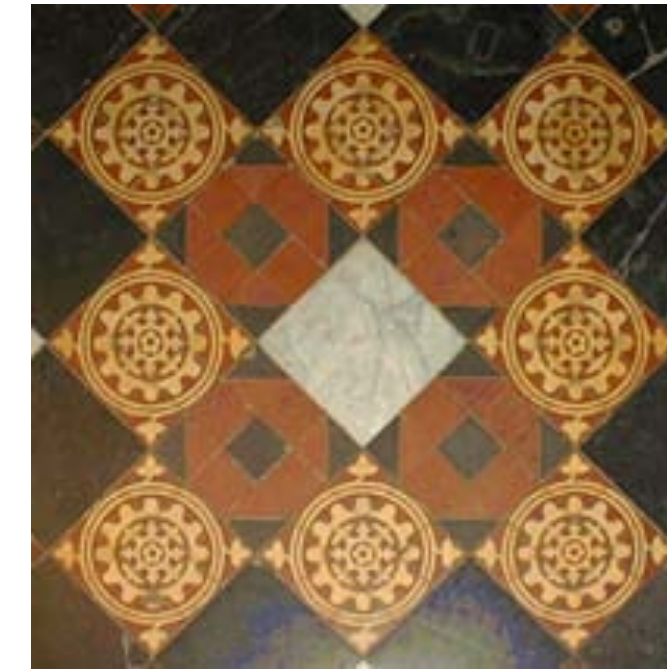
(but what does parallelism of lines mean in a projective space?)

An affine transform preserves $\mathbf{l}_\infty$

A hierarchy of transformations: Affine



$$\mathbf{x}' = \mathbf{H}_A \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \mathbf{x}$$



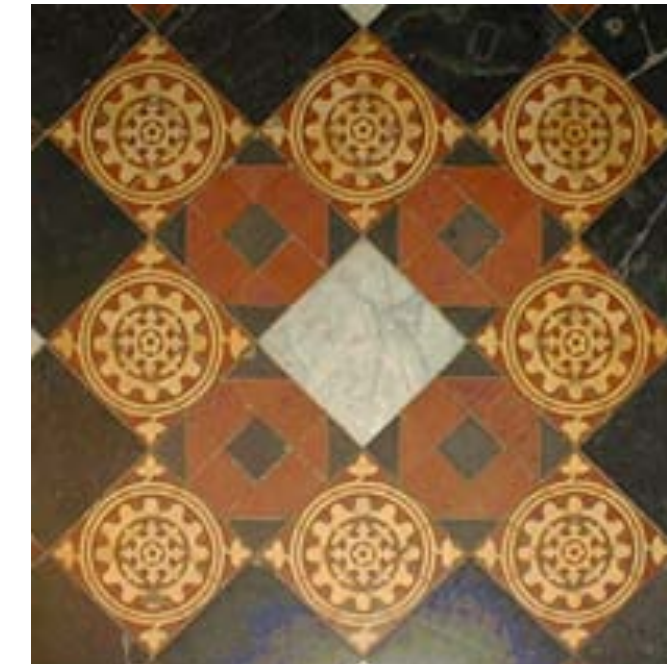
Exercise: show that the line at infinity is preserved if and only if $\mathbf{H}$ is an affine transform

An affine transform preserves $\mathbf{l}_\infty$

A hierarchy of transformations: Projective



$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$



$$\mathbf{H} = \mathbf{H}_S \mathbf{H}_A \mathbf{H}_P = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{v}^T & v \end{bmatrix}$$

A general projective transform can be decomposed into a similarity transform, and ‘minimal’ affine and projective transforms

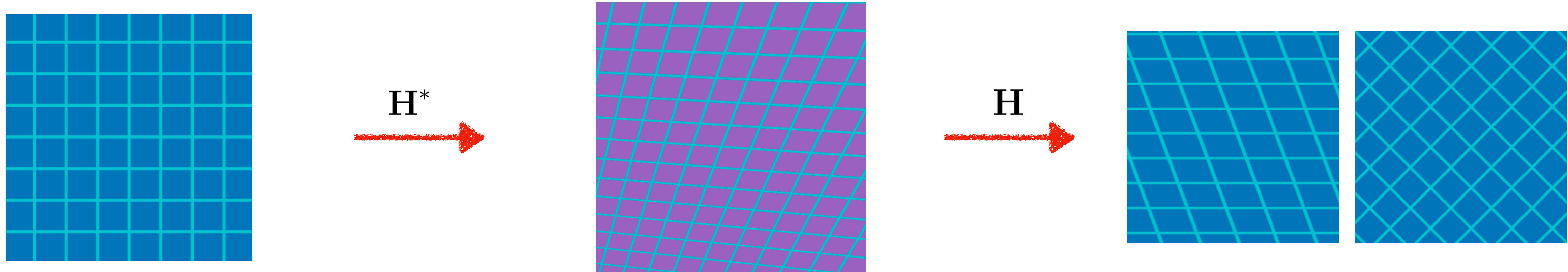
Also possible in reverse order (for different values of s , $\mathbf{v}$, $\mathbf{R}$, $\mathbf{K}$ etc.)

$$\mathbf{H} = \mathbf{H}_P \mathbf{H}_A \mathbf{H}_S = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{v}^T & 1 \end{bmatrix} \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

Q: Why 1 instead of v ?

A: $s, \mathbf{t}, v$ can be scaled together to get same $\mathbf{H}$

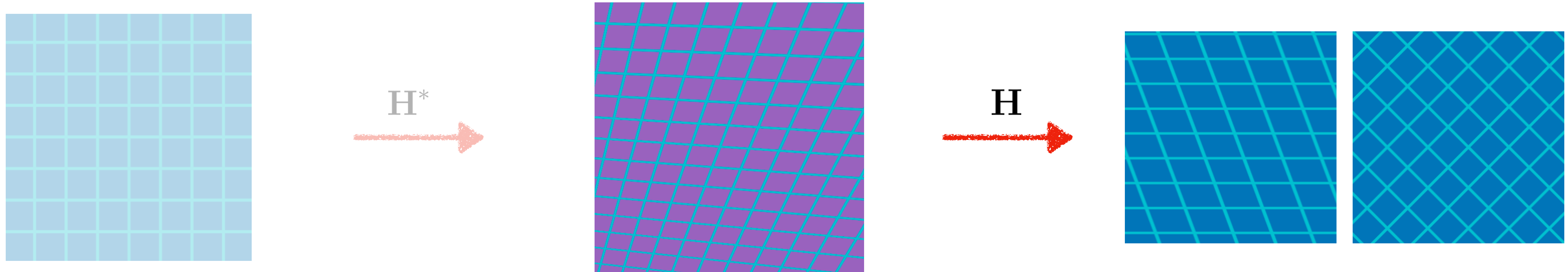
Rectification



A general projective transform does not preserve parallelism/angles

Goal: find **H** that ‘restores’ parallelism (affine rectification) or angles (metric rectification)

Rectification



A general projective transform does not preserve parallelism/angles

Goal: find **H** that ‘restores’ parallelism (affine rectification) or angles (metric rectification)
(without access to **H**^{*})

Rectification



Goal: find **H** that ‘restores’ parallelism (affine rectification) or angles (metric rectification)

A mathematically ill-posed task without additional information
(e.g. which lines should be parallel)

Need user inputs to identify what to ‘restore’

Affine Rectification: Making Parallel lines Parallel Again



Goal: find $\mathbf{H}$ that ‘restores’ parallelism (undoes $\mathbf{H}^*$ upto an affine transform)

Q: How many constraints do we need? i.e. How many parameters to infer?

Any suggestions on the approach?

Hint: what structure identifies parallelism?

Affine Rectification: Making Parallel lines Parallel Again



or

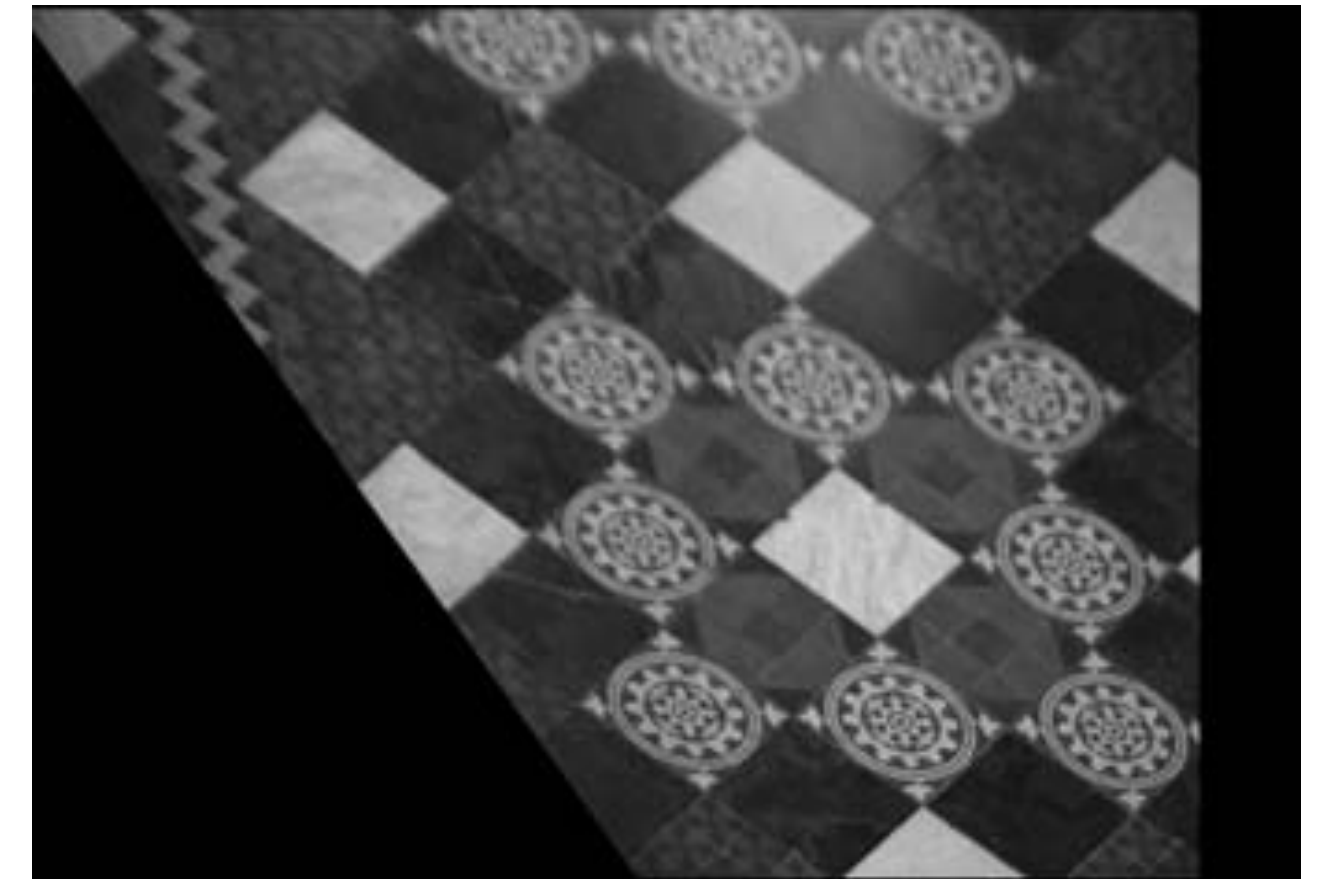
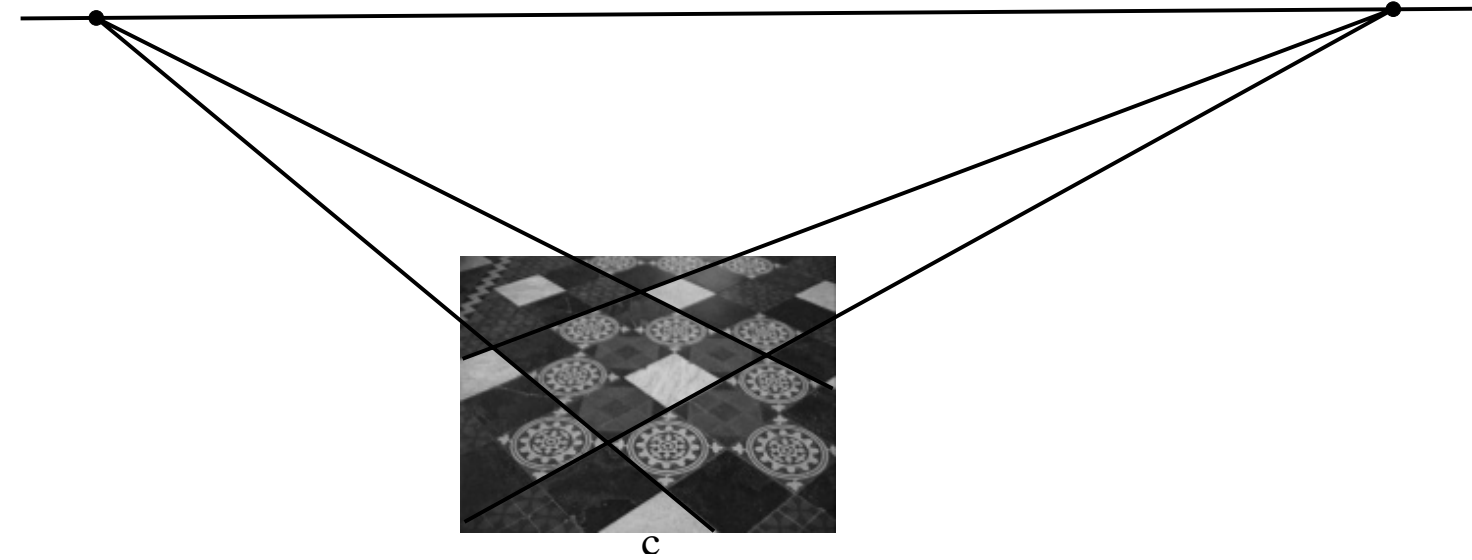


Goal: find $\mathbf{H}$ that ‘restores’ parallelism (undoes $\mathbf{H}^*$ upto an affine transform)

a) Find $\mathbf{l}'_{\infty}$ (i.e. image of the line at infinity in current space)

b) Compute any $\mathbf{H}$ s.t. $\mathbf{l}_{\infty} = \mathbf{H}^{-\top} \mathbf{l}'_{\infty}$

Affine Rectification: Making Parallel lines Parallel Again



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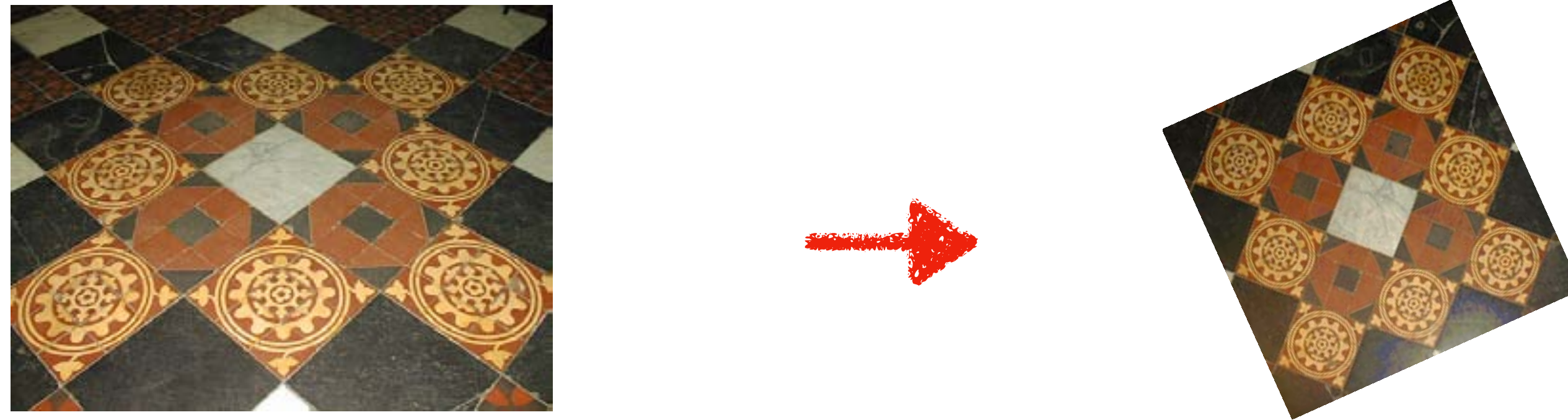
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$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_1 & l_2 & l_3 \end{bmatrix} \quad \mathbf{H} = \mathbf{H}_A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_1 & l_2 & l_3 \end{bmatrix}$$

Metric Rectification: Recovery upto Similarity Transforms



Goal: find $\mathbf{H}$ that ‘restores’ angles (undoes $\mathbf{H}^*$ upto a similarity transform)

Q: For affine transforms, we used the ‘image’ of the line at infinity. Can we do something similar?

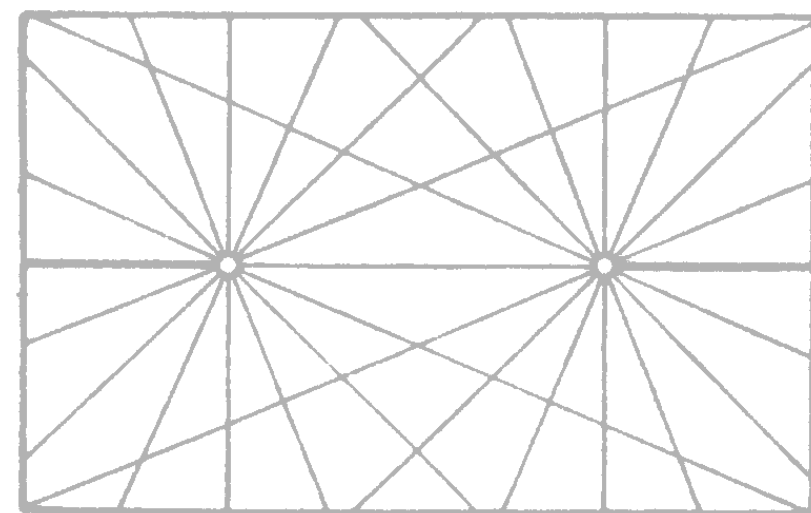
Metric Rectification: Recovery upto Similarity Transforms

$$\mathbf{x}^T \mathbf{C} \mathbf{x} = 0 \quad \text{point conic}$$

$$\mathbf{1}^T \mathbf{C}^* \mathbf{1} = 0 \quad \text{line conic}$$

Example of degenerate line conic: all lines passing through points $\mathbf{x}, \mathbf{y}$

$$\mathbf{C}^* = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$$



Equation of a circle:

$$ax^2 + ay^2 + cxw + dyw + ew^2 = 0$$

All circles pass through 2 ‘circular’ points:
(pretty weird to think about, but easy to verify!)

$$\mathbf{I} = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \quad \mathbf{J} = \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}$$

Circular points are preserved $\longleftrightarrow$ similarity transform

$$\begin{aligned} \mathbf{I}' &= \mathbf{H}_s \mathbf{I} \\ &= \begin{bmatrix} s \cos \theta & -s \sin \theta & t_x \\ s \sin \theta & s \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \\ &= s e^{-i\theta} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \mathbf{I} \end{aligned}$$

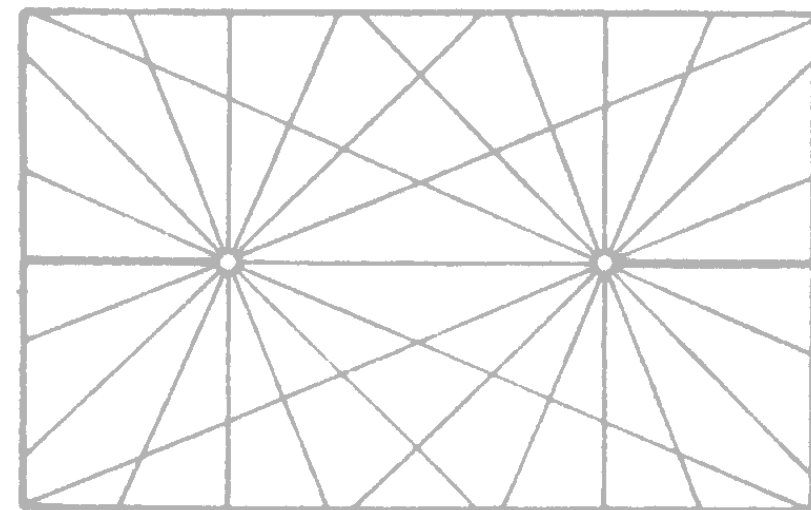
Metric Rectification: Recovery upto Similarity Transforms

$$\mathbf{x}^T \mathbf{C} \mathbf{x} = 0 \quad \text{point conic}$$

$$\mathbf{l}^T \mathbf{C}^* \mathbf{l} = 0 \quad \text{line conic}$$

Example of degenerate line conic: all lines passing through points $\mathbf{x}, \mathbf{y}$

$$\mathbf{C}^* = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$$



Circular points are preserved $\longleftrightarrow$ similarity transform

but, cannot 'image' the points (as they are imaginary!)

Conic dual to circular points:

$$\mathbf{C}_{\infty}^* = \mathbf{I}\mathbf{J}^T + \mathbf{J}\mathbf{I}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

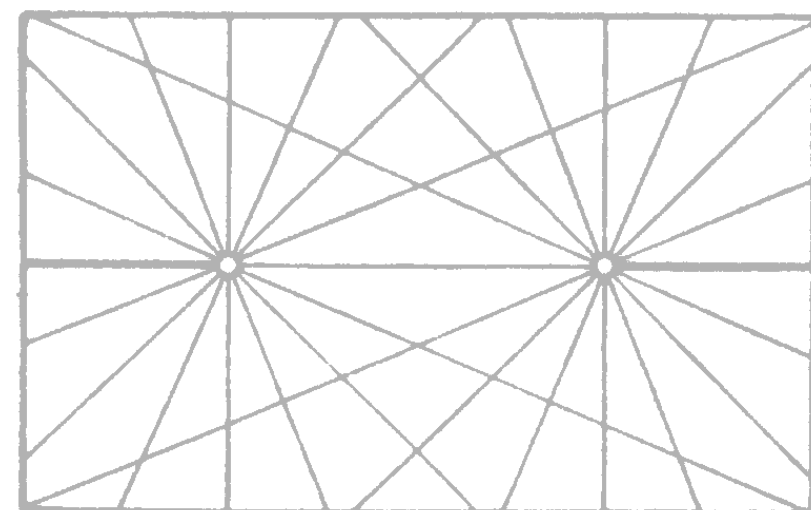
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Angle between 2 lines (in euclidean space):

$$\mathbf{l} = (l_1, l_2, l_3)^T \quad \mathbf{m} = (m_1, m_2, m_3)^T$$

$$\cos \theta = \frac{l_1 m_1 + l_2 m_2}{\sqrt{(l_1^2 + l_2^2)(m_1^2 + m_2^2)}} = \frac{\mathbf{l}^T \mathbf{C}_{\infty}^* \mathbf{m}}{\sqrt{(\mathbf{l}^T \mathbf{C}_{\infty}^* \mathbf{l})(\mathbf{m}^T \mathbf{C}_{\infty}^* \mathbf{m})}}$$

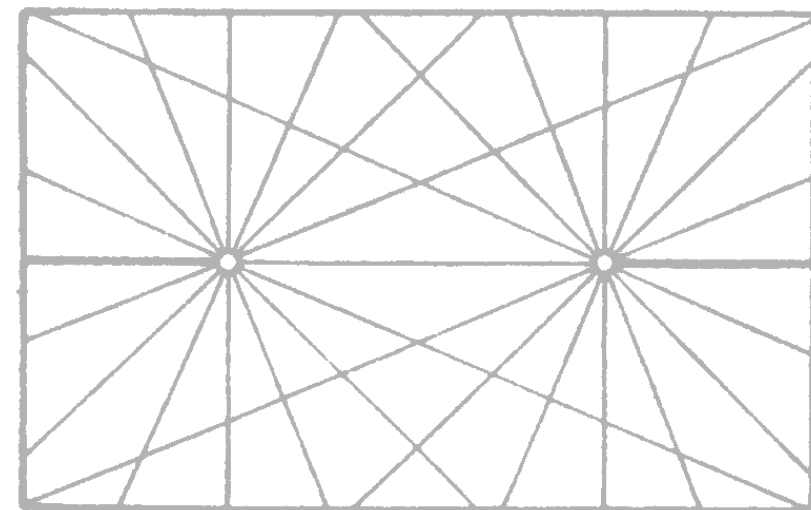
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$$\cos \theta = \frac{\mathbf{l}^T \mathbf{C}_{\infty}^* \mathbf{m}}{\sqrt{(\mathbf{l}^T \mathbf{C}_{\infty}^* \mathbf{l})(\mathbf{m}^T \mathbf{C}_{\infty}^* \mathbf{m})}}$$

Given a homography H:

$$\mathbf{l}' = \mathbf{H}^{-T} \mathbf{l} \quad \mathbf{m}' = \mathbf{H}^{-T} \mathbf{m} \quad \mathbf{C}_{\infty}^{*'} = \mathbf{H} \mathbf{C}_{\infty}^* \mathbf{H}^T$$

$$\cos \theta = \frac{\mathbf{l}'^T \mathbf{C}_{\infty}^{*'} \mathbf{m}'}{\sqrt{(\cdots)} \sqrt{(\cdots)}}$$

In particular, if we know $\mathbf{l}'$ and $\mathbf{m}'$ should be perpendicular,

$$\mathbf{l}'^T \mathbf{C}_{\infty}^{*'} \mathbf{m}' = 0$$

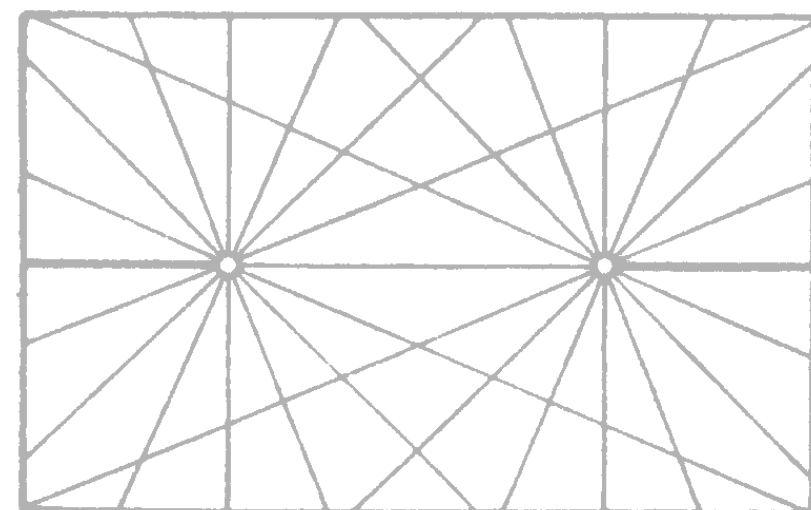
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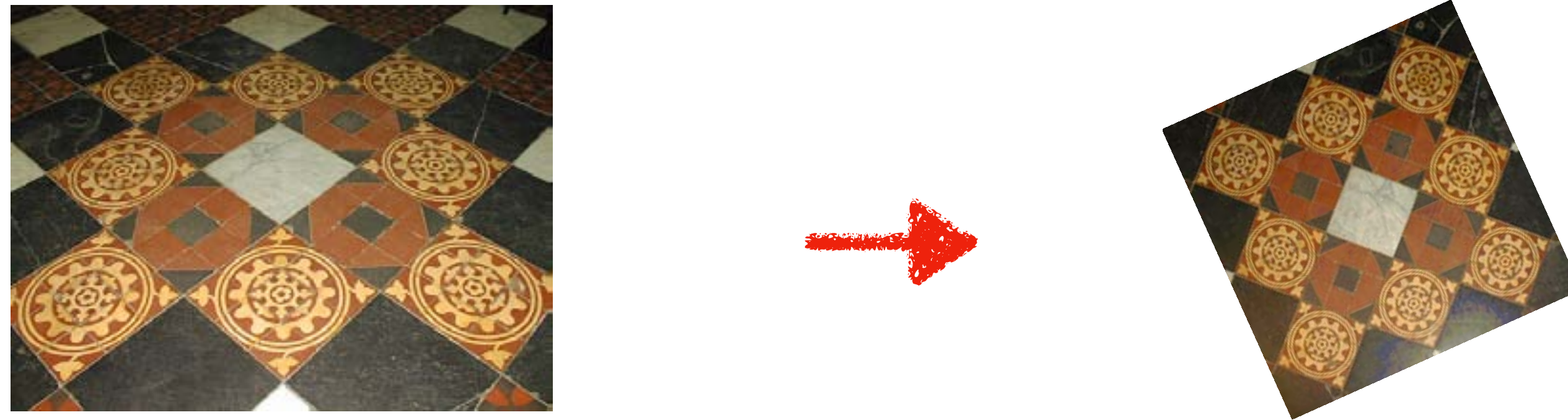
$$\mathbf{C}_{\infty}^* = \mathbf{I}\mathbf{J}^T + \mathbf{J}\mathbf{I}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{C}_{\infty}^{*'} = \mathbf{H} \mathbf{C}_{\infty}^* \mathbf{H}^T$$

In particular, if we know $\mathbf{l}'$ and $\mathbf{m}'$ should be perpendicular,

$$\mathbf{l}'^T \mathbf{C}_{\infty}^{*'} \mathbf{m}' = 0$$

Metric Rectification: Recovery upto Similarity Transforms



Goal: find $\mathbf{H}$ that restores angles (undoes $\mathbf{H}^*$ upto a similarity transform)

- a) Using several perpendicularity constraints, find $C_{\infty}^{* \prime}$
- b) Compute $\mathbf{H}$ s.t. $C_{\infty}^* = \mathbf{H} C_{\infty}^{* \prime} \mathbf{H}^T$

Metric Rectification: Recovery upto Similarity Transforms

$$\mathbf{l}'^T C_{\infty}^{*'} \mathbf{m}' = 0$$

$$C_{\infty}^{*'} = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}$$



If image is already affine-rectified,
then $d=e=f=0$

a) Using several perpendicularity constraints, find $C_{\infty}^{*'}$

$$A\mathbf{c} = 0 \quad A \in \mathbb{R}^{N \times 6}$$

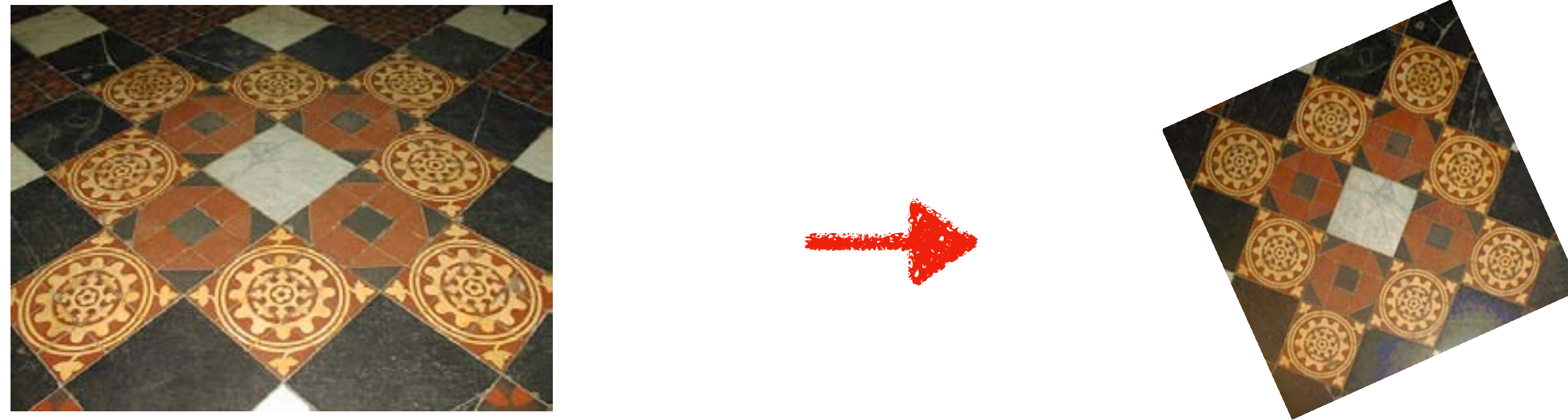
Each pair of perpendicular
lines gives a linear constraint
on the elements!

Q: How many constraints are needed?

≥ 5 in general

≥ 2 if already affine-rectified!

Metric Rectification: Recovery upto Similarity Transforms



b) Compute $\mathbf{H}$ s.t. $C_{\infty}^* = \mathbf{H} C^{*'} \mathbf{H}^T$

$$C_{\infty}^{*'} = U \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T \quad \text{compute SVD!}$$

$$H = \begin{bmatrix} \sqrt{\sigma_1^{-1}} & 0 & 0 \\ 0 & \sqrt{\sigma_2^{-1}} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

Q: Why is SVD of this form?

$$C_{\infty}^{*'} = U D V^T \quad \text{symmetry + rank-2}$$

Note: Possible Bug in H&Z book

Compute $\mathbf{H}$ s.t. $C_{\infty}^* = \mathbf{H} C_{\infty}^{*'} \mathbf{H}^T$

Book Solution:

$$C_{\infty}^{*'} = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T \xrightarrow{\mathbf{H} \text{ s.t.}} HU \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T H^T = C_{\infty}^* \longrightarrow \boxed{H \equiv U^T}$$

~~Compute SVD~~ C_{∞}^*

Possible bug: Not clear why the two singular values are equal, and not —

$$C_{\infty}^{*'} = U \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T$$

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Compute $\mathbf{H}$ s.t. $C_{\infty}^* = \mathbf{H} C_{\infty}^{*'} \mathbf{H}^T$

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C_{∞}^*

Our Solution:

(empirically validated)

$$C_{\infty}^* = U \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T \xrightarrow[\mathbf{H} \text{ s.t.}]{\text{Compute SVD}} HU \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T H^T = C_{\infty}^* \longrightarrow \boxed{H = \begin{bmatrix} \sqrt{\sigma_1^{-1}} & 0 & 0 \\ 0 & \sqrt{\sigma_2^{-1}} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T}$$

C_{∞}^*

Metric Rectification: Recovery upto Similarity Transforms



a



b

Questions?